A Machine Vision-Based Broiler Chicken Weight Detection System Using Deep Learning with Integrated Automatic Feed Regulation

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***Abstract*—Implementation of a Machine Vision-Based Broiler's Weight Detector with integrated automatic feed regulation for the improvement of poultry farms is the aim of the research, as the conventional way of weighing is time-consuming and is a stress ignition for both the farmers and the chickens. The researchers gathered 1,236 images that will cover broiler chickens' weight data from day 1 to day 37. The data set was expanded for data augmentation producing 12,360 photos. The researchers then tested 7 models which are MobileNet, MobileNetV2, InceptionV3, ResNet50, VGG19-based, VGG16-based, and Inception ResNetV2, using metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R2) scores. InceptionResNetV2 model provided the best result with R2 score of 1.00 and MAE of 10.08. While ResNet50 model showed the poorest performance with MAE of 108.76 and R2 score of 0.89. After deciding the best model, it is implemented on the device with a user-friendly graphic user interface. To evaluate the performance of the device, 15 trials were performed on different stages of chickens' growth. The device resulted with an average overall error of 2.505%, average absolute error of 2.668 grams in weight prediction, and an error of 2.005 grams in feed dispensing.**

Keywords—broiler chicken, deep learning, feed regulator, machine vision, weight detector

I. Introduction

Broiler Chickens are chickens bred and raised in captivity purely for meat production. The majority of commercially raised broiler chickens attain their slaughter weight within four to six weeks, while breeds that grow at a slower pace reach this weight around 14 weeks of age. Standard broilers typically display white plumage and possess yellowish skin. Assessing the body weight of Broiler Chickens is a fundamental aspect for most poultry farmers. This metric is significant because it directly influences predicting the readiness of chickens for harvest. The traditional method of weighing is stressful for both the chicken and the farmer. Stress induced by handling can negatively impact the quality of the broilers. Many environmental stresses, including managing, immunization, feed interruptions, raising on used litter, and high room temperatures, can cause stress in poultry. Stressful environments can cause birds to lose their appetite, become lethargic, gain less weight, become less efficient at digesting their diet, and increases the chance of dying [1]. Manual inspection and weighing are also prone to human error, affecting the accuracy of the necessary data. Human vision and other human errors can occur when examining and weighing the broilers. That is why numerous industries use computer vision, including autonomous vehicles, robotics, surveillance, and augmented reality [2]. Image-based research has recently emerged as an achievable option in a variety of fields, including agricultural surveying, human-machine-based systems, and autonomous vehicle navigation, and livestock, among others [3]. Computer vision also serves as a non-invasive technique that has been proven and used to monitor animals' features, behaviors, and locations. It can also help in enhancing farmers' efficiency which can lead to increased production. Additionally, these advancements can offer remote assistance, potentially reducing their exposure to environmental elements and lowering the risk of health complications. Computer vision has also been used in determining the weight of pigs by capturing images of their back. They use multiple output regression CNN and estimate the weight based on the pig's height, contour, and body shape. However, they can only measure pigs individually [4]. Another application is the use of computer vision to determine the weight of fish. With the use of Linear Regression, Random Forest Regression, and support vector regression, researchers were able to identify the weight of tilapia with an R2 value of 0.70. One limitation of this study is that it requires prior knowledge of the fish's age [5]. Another related study is the weight estimation on Pekin ducks slaughter factory with the ducks on a routine movement [20]. A similar study on chickens weight estimation was also conducted but focused on older chickens. Related researches suggested that the weight estimation error increases as the chickens grow older, thus gives the motivation of this project to gather data set as the broiler chicken aged [21]. Developing a system that uses machine vision and AI to detect broiler chicken weights, combined with an automatic feeder that adjusts feed amounts based on those weights, can revolutionize poultry farming operations. This integrated approach enables accurate weight tracking, optimized feed delivery based on the flock's needs,

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consistent growth rate and harvest weight, and more sustainable broiler chicken production overall.

II. Related works

Broiler Chickens are one of the most common providers of meat in many countries in the world. Reference [6] stated that broiler chickens provide efficient feed conversion to meat. The estimated annual global production is more than 90 million tons of meat, proving that broiler chicken is one of the main food sources [7]. This type of chicken contributes to the success of the economy of most poultry, it is a reason why proper management is required in this type of industry. These types of poultry are also supplying eggs to contribute in terms of meeting the national protein supply [8]. As per reference [9], modern broiler chickens differ from domestic chickens raised before the middle of the twentieth century in terms of physical features, genetics, and isotope. Modern broilers grow more quickly than older breeds, which has greatly increased their energy efficiency because they reach slaughter weight faster and require less energy for physical activity and other metabolic heat-producing processes like protein turnover [10] [11]. Reference [7] state that the traditional weighing of broiler chickens has two types which are the manual and automatic platform weighing systems. The manual platform weighing system is the oldest way of getting the broiler chickens' weight which will require much human labor due to the requirement of catching the chicken and putting it in the platform which will also require much time, this may also cause stress to the chickens. The automatic platform weighing system is better regarding the chickens' health and human labor. In this method of weighing system, the chickens are weighed once they step on the specific spot of the poultry voluntarily wherein the platform is located, the only problem is some heavier broilers will visit the platform rarely [12] [13]. The broilers' body weight plays an important role in their production since it expresses the health of the flock and the efficiency of production [14] [15]. It is also mentioned by reference [16] that a quick assessment of how feed distribution impacts body weight growth can be obtained by weighing broilers more frequently than once per week, weight is an essential indicator of a broiler chicken's health during the rearing process which is a key factor affecting the quality features of meat [17]. The spot where the cameras are placed is at the top wherein it can get an aerial-view image of the subject, as mentioned in reference [18], it is to determine body surface dimensions and may be used for real-time monitoring. It has been suggested that visual image analysis (VIA) be used to continuously monitor the weight increase performance of pigs, enabling earlier problem diagnosis and better management choices. However, it could only monitor the weight of the pigs individually and future work could focus on the effect of pig's gender and genotype on body surface area and body weight. [18] As indicated by reference [19], infrared-light depth sensors may be added to the VIA due to their functionality in avoiding errors related to the subject's color and inaccuracies caused by the variation of surroundings that can affect the visible light in the VIA. One such related work involves estimating the weight of white Pekin ducks using computer vision, but unlike with this research wherein alive chickens will be monitored, their samples where duck carcasses on a production line of slaughter factories making them static and easier to capture in images [20]. Another related work is almost similar to this project, using image processing to measure the weight of broilers, but can only weigh the broiler chicken one by one which is very time consuming when implementing it in a poultry farm. The study focused on measuring the weight of older chickens, and it mentioned lacking data set on broiler chicken of significant age. Thus, the study recommended considering in the future having image datasets for a variety of day age groups of broiler chickens which is the motivation of this research to gather datasets from chicks to harvest ready broiler chickens [21]. Aside from determining the weight of the chickens the project will also correlate the measured weight to the feed amount and incorporate an automatic feeding system. Reference [22] describes the creation of an automated chicken feeding system utilizing Raspberry Pi components to improve the efficiency of poultry farming. This system automates the processes of feeding, counting, and storing food, while also monitoring the freshness of the food and ambient conditions. The idea is designed to tackle labor shortages, reduce food waste, and promote chicken health, thereby enhancing sustainable poultry production. While reference [23] presents an automated feeder for laying hens, incorporating Arduino Uno and real-time clock (RTC) technology. This technique allows for accurate and timely feeding management, improving efficiency and effectiveness when compared to manual feeding approaches. The results show minimal mistake rates, emphasizing the feeder's dependability in delivering precise feeding regimens for laying hens. Each broiler chicken will require about 1 ounce per day or approximately one pound of feed for the first 14 days [24]. Reference [25] utilizes RTC module. The study creates an automated system that controls and monitors temperature in chicken coops by utilizing a microcontroller. The system utilizes a DS3231 RTC module and servo motor for feeding, together with a DHT22 sensor and bulb for temperature monitoring. Feeding is scheduled at specific times, while the lamp automatically changes according to temperature conditions. The device offers automated monitoring of food and cage temperature, providing simplicity and efficiency for chicken farmers. As per reference [26], automated chicken feeding reduces labor shortages in chicken farms, and it guarantees poultry receive appropriate nutrition and minimizes labor expenses. Although automated chicken coop has been developed, the project proposes a more feasible system wherein the feeds will be regulated based on the measured real-time broiler chicken weight. Limitations from previous studies that will be addressed by this project are that it will measure alive and moving group of chickens, without prior knowledge on the initial age of the chickens, and no need to have the chickens go to a designated platform with weight can be measured as frequent as possible. Because the project will gather data set from day 1 chicken as they grow to harvest ready condition and that the data gathering procedures will be done with flocks of broiler chickens. Unlike with other automated feeding system wherein same feed amount will be given on the scheduled time, the project will vary the given feeds based on the measured broiler chicken weights.

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III. Research Methodology

This research is dedicated to developing a prototype that enhances the traditional system for weighing broiler chickens with a feeder system that tracks the amount of chicken feeds that will be given based on their weights. Through a series of tests, the researchers hope to gather data for designing an improved system. The overview of the proposed work is presented on Fig. 1. For 37 days, 30 images of broiler chickens made up the original dataset. Each chicken is manually cropped, creating a total of 1,236 image data as shown in Fig. 2 and Fig. 3. Every picture was correlated with the weight of the chicken, creating a large set of data that could be used to train and evaluate machine learning models.

A. Data Augmentation and Image Preprocessing

The dataset was expanded for data augmentation by augmenting each image 10 times. This produced 12,360 photos after adding artificial data. By preparing the models for potential sudden shifts in angles, orientation, distance, and lighting, image processing increased the models' resistance to variations in image capture settings. A key component of this process was the application of gray scaling the images using OpenCV, Gaussian blur for smoothing image and using OTSU method that segment the images into binary as shown in Fig. 4. The Gaussian blur helped to reduce image noise and detail, making it easier to isolate the chicken shapes from their surroundings. This segmentation is crucial for accurate weight estimation, as it allows the model to focus solely on the relevant features of the chickens. The image was rotated by a random angle between -10 and 10 degrees. In this case, the image was horizontally flipped (there is a 50% probability of the image to be flipped). The image was then scaled by a random factor between 0.9 and 1.1. These augmentations help the model learn to recognize chickens in various orientations and positions, improving its ability to generalize to real-world scenarios where chickens may be captured from different angles. By artificially increasing the diversity of the training data, these augmentation techniques likely contributed to the model's robust performance across different chicken sizes and positions.

B. Deep Learning Modeling

Seven different modeling techniques were used by the researchers, and each was configured to run for 50 epochs with a batch size of 16. MobileNet and MobileNetV2 are chosen because they employ a unique method known as “depthwise separable convolutions”, which lowers the amount of computing required. Neural networks VGG 16 and VGG 19 are renowned for being straightforward and reliable. All of their layers use tiny 3x3 filters, which make them simple to comprehend and apply. While ResNet50 is a promising option because of its ability to recognize intricate patterns in photos. Lastly, InceptionV3 uses filters of various sizes inside the same layer to examine images in great detail. And Inception-ResNet-v2 combines the features of ResNet50 and the InceptionV3.

- MobileNetV2-based model: For effective real-time processing, such as object identification for estimating the weight of broiler chicks, the MobileNetV2 model is employed. Importing libraries like numpy, matplotlib, tensorflow, and keras is the first step in the analytical procedure. Data preparation, which involves preprocessing images and dividing the dataset into training and testing sets, comes next. After loading the pre-trained MobileNetV2 model, further layers are added to the top of the base model. The model is then trained to fit the training set of data. When the model has been tested on the test set, it is prepared to generate precise predictions.

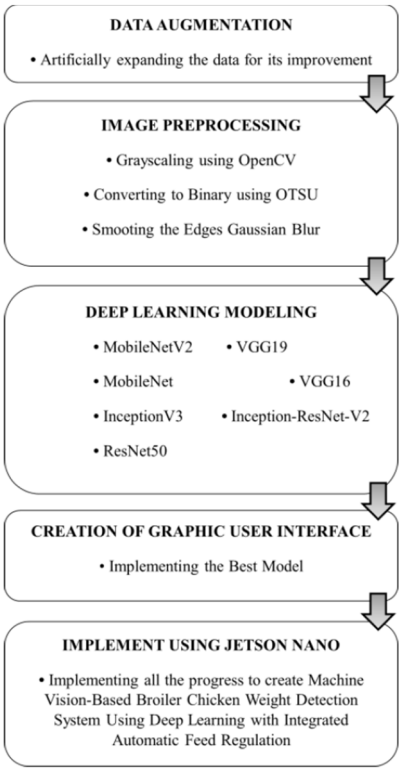


Fig. 1. Overview of Proposed Work (Methodology Framework)



Fig. 2. Original Image



Fig. 3. Image Data Cropping



Fig. 4. Image Segmentation with Gaussian Blur

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- MobileNet-based model: The MobileNet model is used for effective real-time processing, just like MobileNetV2. The process begins with the import of the required libraries, followed by dataset separation and picture preprocessing to prepare the data. Custom layers are added to the MobileNet model to improve it after it has been pre-trained. After that, the model is assessed on the test set, trained on the training data, and used to forecast the weight of the broiler chicks
- InceptionV3-based model: The InceptionV3 model is known for its effectiveness in managing complex models and huge datasets. Importing libraries and getting the data ready are the first steps in the process. After loading the pre-trained InceptionV3 model, custom layers are added. Metrics like mean absolute error (MAE), mean square error (MSE), and root mean square error (RMSE) are used to assess the model after it has been trained. Lastly, the weight of the broiler birds is accurately predicted by the algorithm.
- ResNet50-based model: ResNet50 is a 50-layer deep network designed to solve vanishing gradient issues by utilizing residual blocks. Importing libraries, preparing the data, and launching the 50-layered deep convolutional neural network are all part of the methodical process. The rectified linear unit (ReLU) is used for non-linear transformations in the model, and a global average pooling layer is used to manage the dimensionality of the feature maps. The last stage is to assess the model and generate precise estimates for the weight of the broiler chicks.
- VGG19-based model: A deep neural network with nineteen layers, the VGG19 model consists of convolutional, max-pooling, fully connected, and softmax layers. Importing libraries is the first step in the procedure, and then data preparation. The input layers are the first in the model, and are followed by convolutional, max-pooling, fully connected, and output softmax layers. Metrics like MAE, MSE, and RMSE are then used to train and assess the model.
- VGG16-based model: Thirteen convolutional layers and three fully linked layers make up the sixteen layers of the VGG16 model. Importing libraries and getting the data available come first in the process. With a single neuron in the output layer that uses a linear function, the sixteen-layer deep network model is suited for estimating the weight of broiler chicken. The model's hyperparameters are optimized using the Keras Tuner, and then the model is trained, its accuracy and performance assessed, and predictions are made.
- Inception-ResNet-V2-based model: The advantages of the ResNet and Inception architectures are combined in the Inception-ResNet-V2 model. Importing libraries, prepping the data, and loading the Inception-ResNet-V2 model that has already been trained are the steps in the procedure. The base model is topped with custom layers. The training data is used to train the model, and relevant metrics are used to assess its performance. Lastly, it accurately forecasts the weight of broiler birds.

C. Creation of the Graphic User Interface and Implement Using Jetson Nano

The most accurate deep learning model was first integrated into the feeder's control system in order to implement the automatic feeder regulation. Using processed images as a basis, this algorithm was utilized to predict the weight of broiler chickens in real time. The feed regulation was then controlled by an algorithm created using the average weights of the chicken with Table 1 as the basis for the chicken's weight per number of feeds needed. This program determined the flock's average weight, compared it to a broiler chicken ideal weight curve, and changed feed levels that was based by the reference [27] to account for underweight or overweight flock members by using Arduino as the feeder's main control panel in the stepper motor. The chickens' estimated weights and growth stages were used to determine the best times to dispense feed according to an automated feeding schedule by the users. This comprehensive approach made sure the automatic feeder could adjust to the flock's changing specifications which might enhance feed efficiency and flock health in general. The integration of the weight prediction model with an automatic feeder system allows for precise feed regulation based on the estimated weights of the broiler chickens. This integration can optimize feed efficiency, improve growth rates, and enhance production in broiler chicken farms. Shown in Table 1 is the established suggested grams of feed intake per day per chicken to correlate the feed intake per day to the predicted weight [27]. The graphic user interface displayed necessities, such as the device's purpose and the start button on the first page, so that the user could visualize how to use it. It then led the user to the screen where the camera input and the estimated weight of the broiler chicken were displayed to directly show the device's output. The study assesses the weight difference between manually and autonomously reared broiler chickens, the efficacy of the weighing system, and the precision of the feeds that will be given based by the weight of the chickens' mechanism at different time intervals.

IV. Results and Discussion

A. Comparison of Deep Learning Models Performance

The researchers utilized seven models to assess which would provide the most effective model in predicting the weight of broiler chickens.

TABLE I. Feed intake of chickens based on weight

Grams of Chicken	Grams of feed needed
0 grams – 369 grams	35 grams
369 grams – 809 grams	87 grams
809 grams – 1352 grams	120 grams
1352 grams – 1966 grams	133 grams

The models evaluated were MobileNetV2, MobileNet, InceptionV3, ResNet50, VGG19, VGG16, and Inception-ResNet-V2. Each model was evaluated using metrics including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R2) scores. For Table 2, it is shown that two models achieved to reach a R2 scores of 1.00 making it the best R2

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scores, these models are the MobileNet and the Inception-ResNet-V2. On the other hand, comparing these two models resulted that the Inception-ResNet-V2 model is better due to its lower MAE which is 10.08, compared to the MAE of MobileNet which is 16. 98. The ResNet50 model showed the poorest performance among the tested models, with the highest MAE (108.76) and lowest R2 score (0.89).

B. Integration of the Model to the Hardware Device

The developed Machine Vision-Based Broiler Chicken Weight Detection System with Integrated Automatic Feed Regulation features a user-friendly graphical user interface. The interface as shown in Fig. 5. provides key information such as average chicken weight, suggested feed amount per chicken, and total feed amount for the flock. The user presses the "Show Processed View" button to analyze the flock.

C. Testing the Device Prediction of Chicken Weight and the Amount of Feeds

To evaluate the performance and accuracy of the Machine Vision-Based Broiler Chicken Weight Detection System with Integrated Automatic Feed Regulation, 15 trials were conducted. These series of trials were performed on two groups of chickens: chicks and large chickens with increasing interval of 1 chicken starting from only 1 chicken on the device up to 10 chickens. For each trial, the system's predicted weight was compared to the actual weight, and the amount of feed dispensed by the automatic feeder was recorded.

TABLE II. COMPARISON OF MODEL PERFORMANCE ON TEST SET

Model	MAE	MSE	RMSE	R2
MobileNetV2	70.66	10611.9	103.01	0.95
MobileNet	16.98	526.26	22.94	1.00
InceptionV3	24.62	1216.00	34.87	0.99
ResNet50	108.76	25112.6	158.47	0.89
VGG19	61.1	7659.2	87.52	0.97
VGG16	101.73	19053.61	138.03	0.92
Inception-ResNet-V2	10.08	164.22	12.81	1.00

TABLE III. PERCENTAGE ERROR FOR EACH TRIAL OF CHICKS

Trial Number / No. of Chickens	Error of Predicted Weight	Error of Predicted Number of Chickens	Error of Suggested Feed Amount
1	2.63%	0.00%	3.56%
2	2.60%	0.00%	0.88%
3	2.62%	0.00%	0.32%
4	2.69%	0.00%	1.10%
5	2.73%	5.00%	1.04%
6	2.80%	0.00%	0.99%
7	2.36%	0.00%	0.58%
8	1.97%	0.83%	0.76%
9	2.33%	0.00%	3.33%
10	2.32%	0.67%	0.43%
Average	**2.505%**	**0.65%**	**1.299%**



Fig. 5. Graphic User Interface (Processed View)

Across all the trials, its average actual weight of the chicks reached to a total of 106.5 grams, while the system predictions have ranged from 108.59 – 109.48 grams resulting in an average predicted weight of 109.168 grams. Its average percentage error of predicted weight resulted to a total of 2.505%, error of predicted number of chickens of 0.65%, and error of suggested feed amount of 1.299% as shown in Table 3. The conducted paired t-test and non-inferiority test for the actual and predicted chicken average weight with 150 samples, resulted with a mean difference of -2.668 grams. This indicates a slight overestimation by the system, with an average absolute error of 2.668 grams. While, the conducted paired t-test and non-inferiority test for the actual and recommended feed amount resulted with a mean difference of -2.005 grams. Thus, the automatic feeder dispensed a precise amount according to the actual feed with acquiring an error of only 2.005 grams.

V. CONCLUSION

The study of the Machine Vision-Based Broiler Chicken Weight Detection System with Integrated Automatic Feed Regulation yielded promising results. The Inception-ResNet-V2 model emerged as the superior deep learning model for predicting broiler chicken weights from images, demonstrating perfect R-squared scores and low Mean Absolute Errors. The integrated system showcased high accuracy in weight prediction, with an average error of only 2.5%, and provided precise feed amount recommendations. Statistical analyses further confirmed the system's reliability in both weight estimation and feed amount calculation. The development of a user-friendly Graphical User Interface (GUI) enhanced the system's practical applicability in real-world poultry management scenarios. Overall, the system shows significant potential for improving production efficiency in broiler chicken farming through automated weight estimation and feed regulation. It is advised to implement the system with a 97.5% adjustment to the recommended feed amount to account for the slight overestimation observed. Establishing a feedback loop with periodic manual weight measurements is crucial for fine-tuning the system's accuracy over time. Further research should focus on investigating the long-term impacts on feed conversion ratios and overall flock health, as well as adapting the system for different breeds and growth stages of broiler chickens. Lastly, exploring the potential for scaling up the system for larger commercial poultry operations is recommended, including the development of comprehensive

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training programs for farmers and poultry managers. These recommendations aim to maximize the system's effectiveness and broaden its applicability in the poultry industry.

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